

Olist DW Project

Summary Of the Project Objective:

- Building a Full BI Solution for Olist E-commerce Business to Collect, Transform, Analyzing, Model and Visualize Olist Operational Data to figure out the Company's performance to get an overview insights to enhance their Operations, Sales and Business Using Different Methodologies and Tools.

Business Requirements:

- Define The Scenario of the Business

1. An Order Might Have Multiple Items.
2. An Order Must be Ordered by A Customer
2. Each Item Might be Fulfilled by a Distinct Seller.
3. Each Customer Might Have Multiple Orders
4. Each Customer Might Have one Location
5. Each Order Might Have One Review
6. Each Order Must Have Payment
7. An Item Must Have a Product
8. Each Product Must Have a Category|

Business scenario

- Define the Audience

- Define Business Questions

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1. What are the Products most sold ?
 2. What are the payments most used ?
 3. How many Orders made by a Single Customer ?
 4. How many Items in a Single Order ?
 5. What are The Items most common ?
 6. Who are the Sellers with most Orders ?
 7. Which States contributes the most for each Product ?
 8. Which States contributes the most for each Seller ?
 9. Which States contributes the most for each Payment type ?
 10. Which Products are most Satisfying ?
 11. Which Seller are most Satisfying ?
 12. How to optimize Delivery Time ?

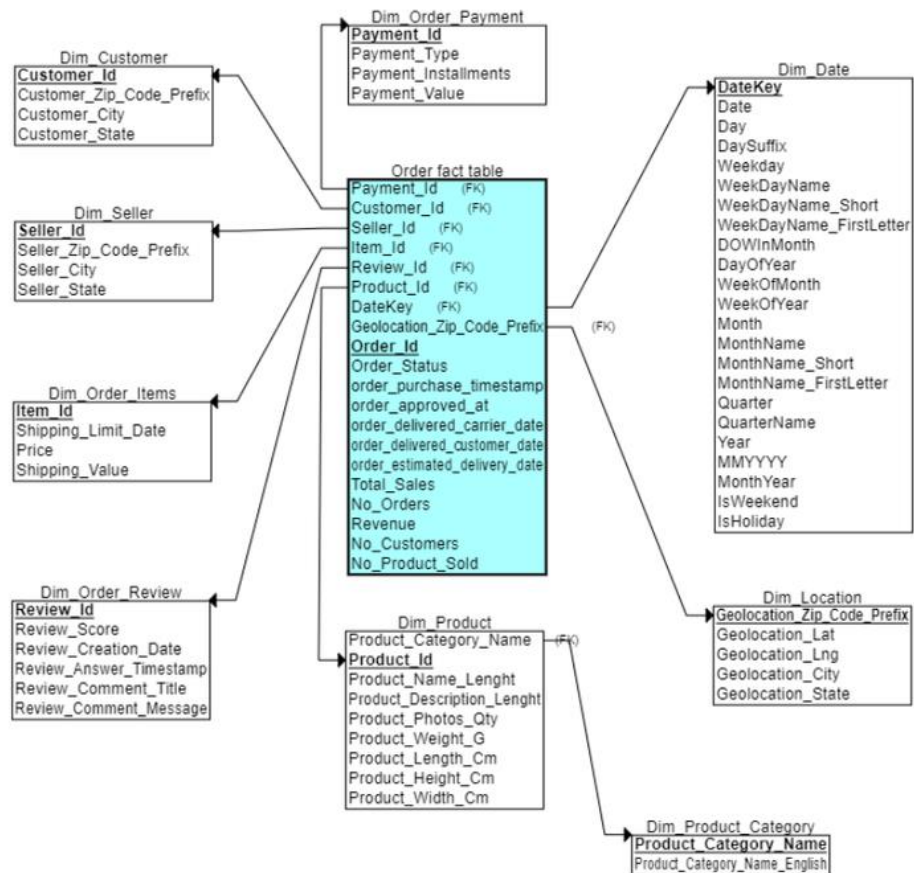
Business Questions

- Going with the Approach of “There is no Return Rule for Orders “

- Going with the Approach of “Keeping the Latest Records”While Updating the Data Warehouse

DW Design:

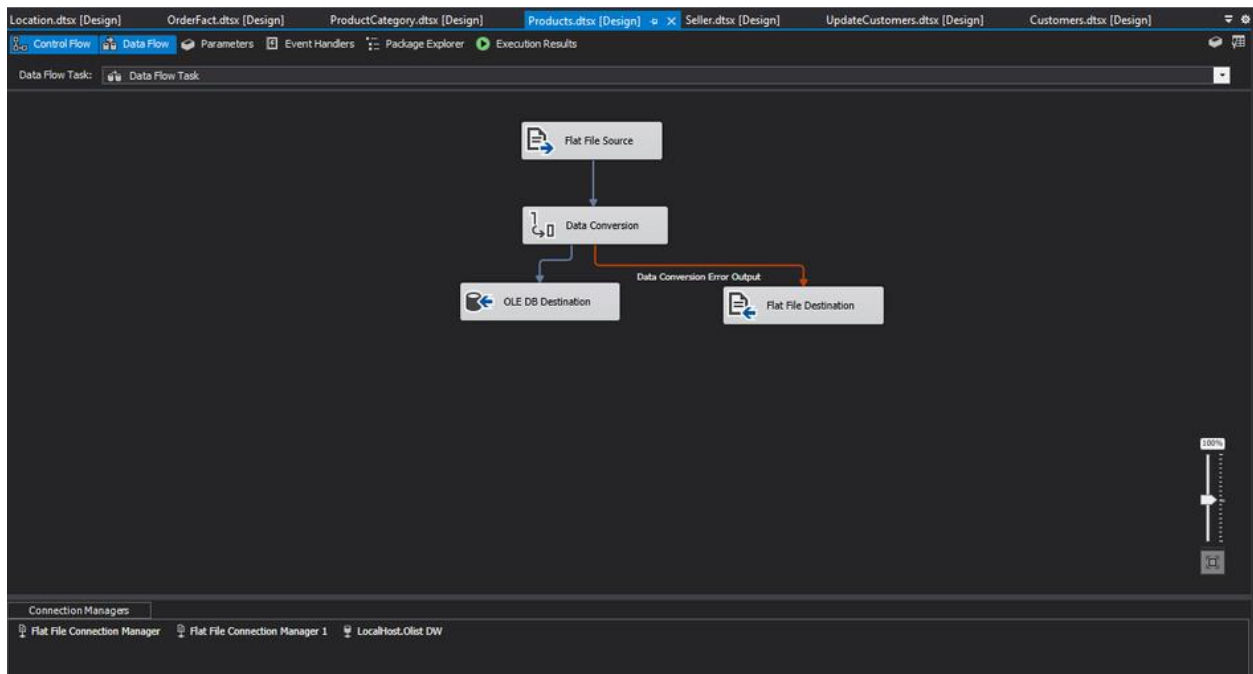
- By Using Dimensional Modeling (Kimball) Methodology, Identified the Fact, The Dimensions, Measures and Aggregation
- Also Used Snowflake Schema in the Design To bring out all the Needed Facts and Dimensions from the Data



Data Warehouse Schema

ETL Process :

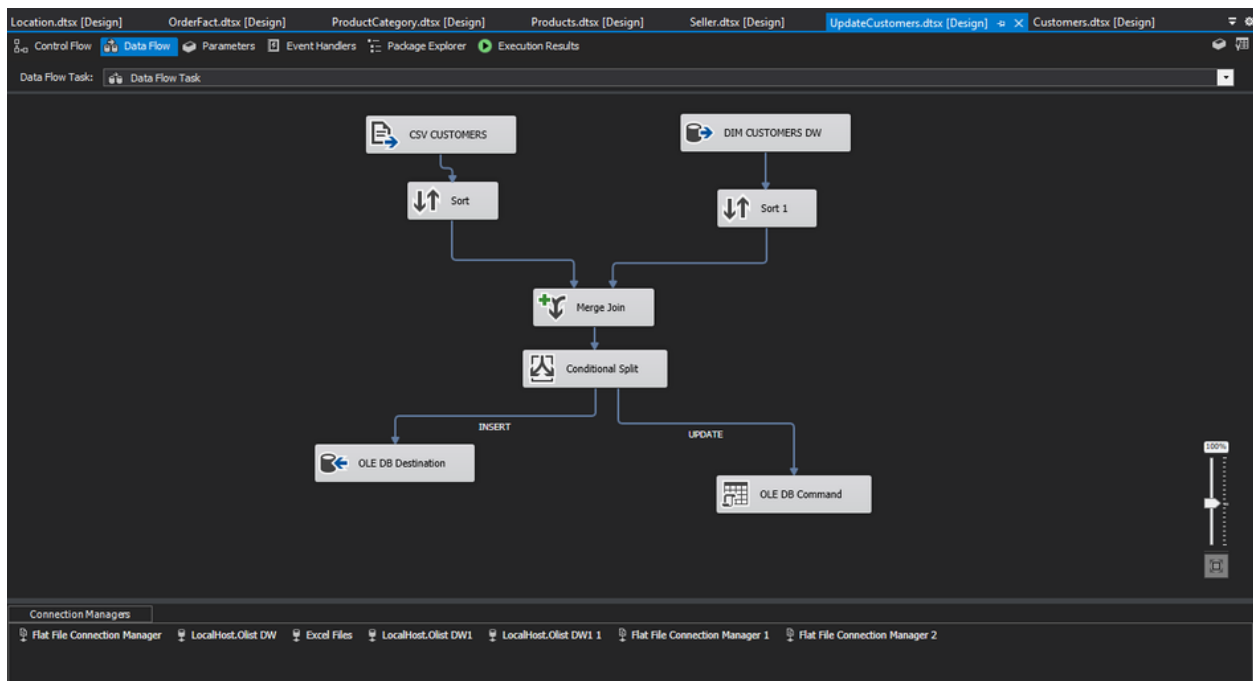
* Filling DW: - Building ETL Packages Using SSIS to Extract all the Data from CSV File and doing all the Transformation needed on the Data to be ready to Load it into the Data Warehouse.



Filling Data

* Updating DW:

- Building ETL Packages Using SSIS To Update the Data Warehouse with the New Data by Getting the Data from CSV Files and doing all the Transformation needed, Then Inserting the New Records and Updating the Old Ones.



Updating Data

Dashboard Creation:

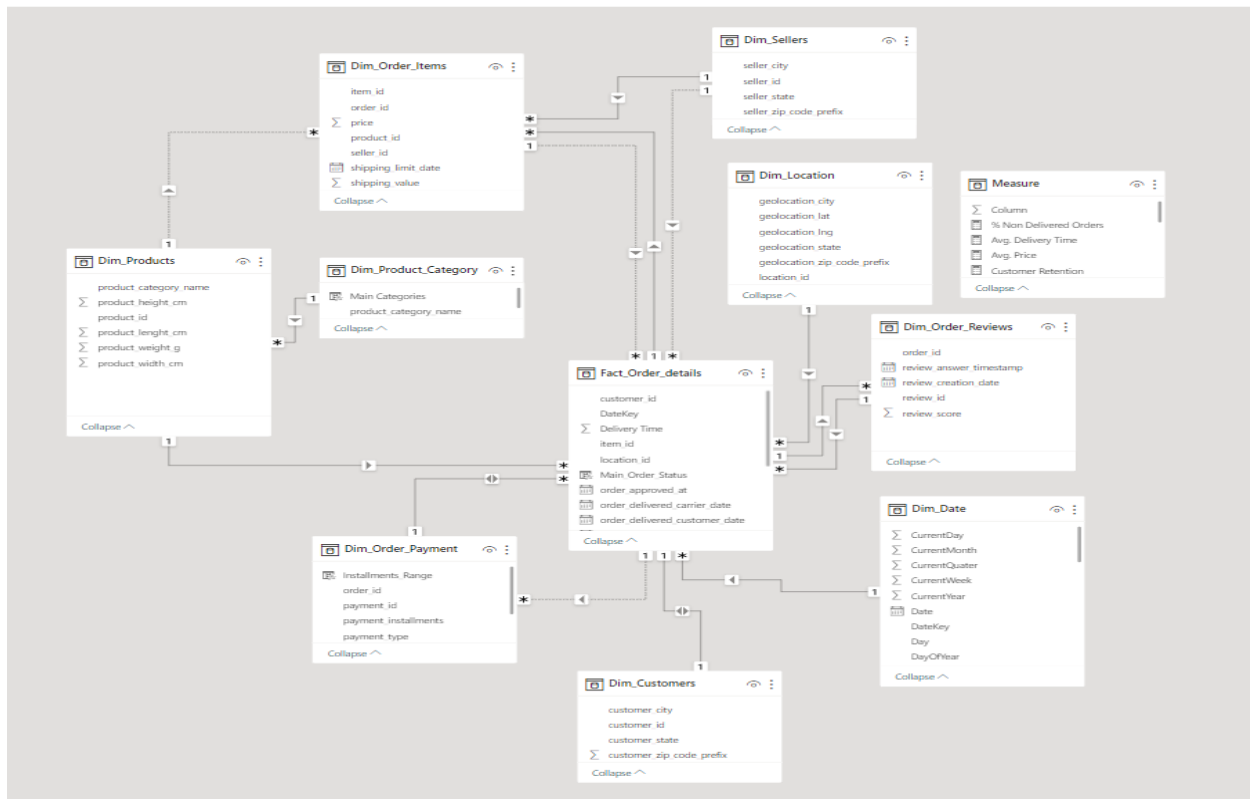
* Data Transformation Using Power Query:

- Changing Data Types
- Removing Unnecessary Columns
- Removing Duplicates
- Adding Custom Columns that benefits the analysis
- Dealing With Null Values Based on Its need

* Measures and Facts:

- Total Revenue
- Score Rate
- Avg. Price
- Avg. Delivery Time
- % of Non Delivered Orders
- No. Of Delivered Orders
- No. Of Orders
- No. Of Customers
- Customer Retention
- No. Of Products
- No. Of Products Sold
- Total Revenue Last Year

* Data Modeling:



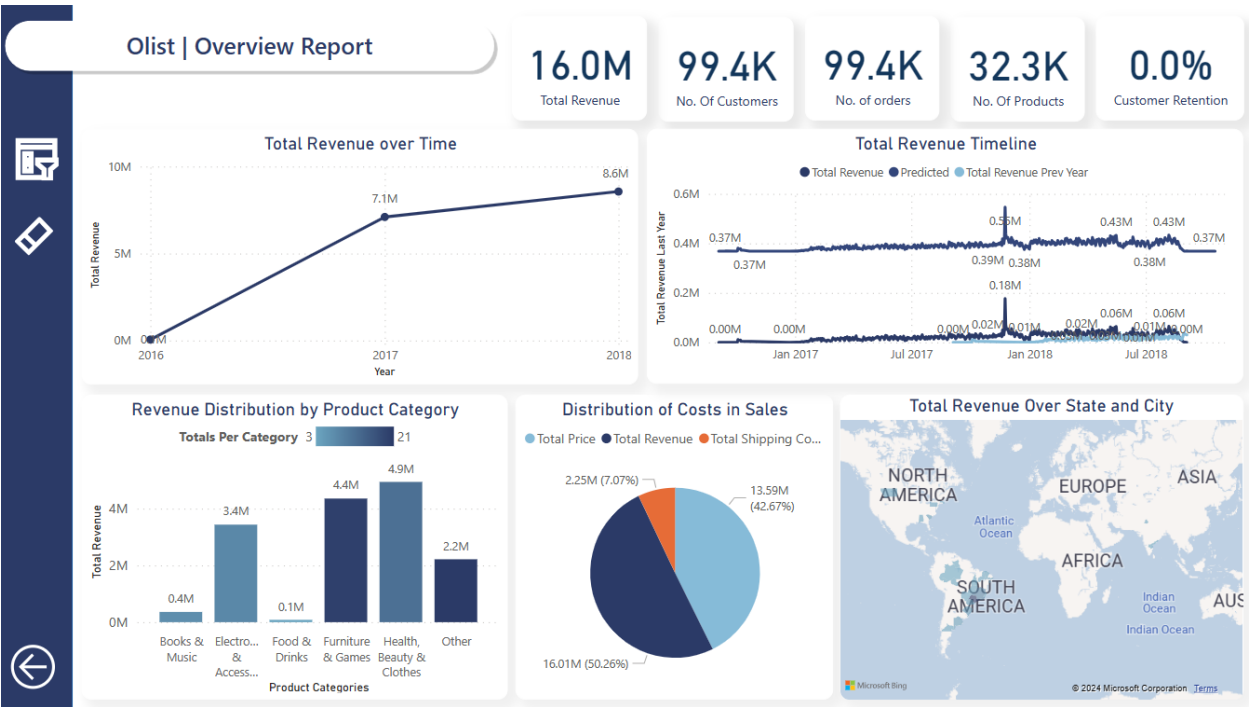
Data Modelling

* Visualizations:

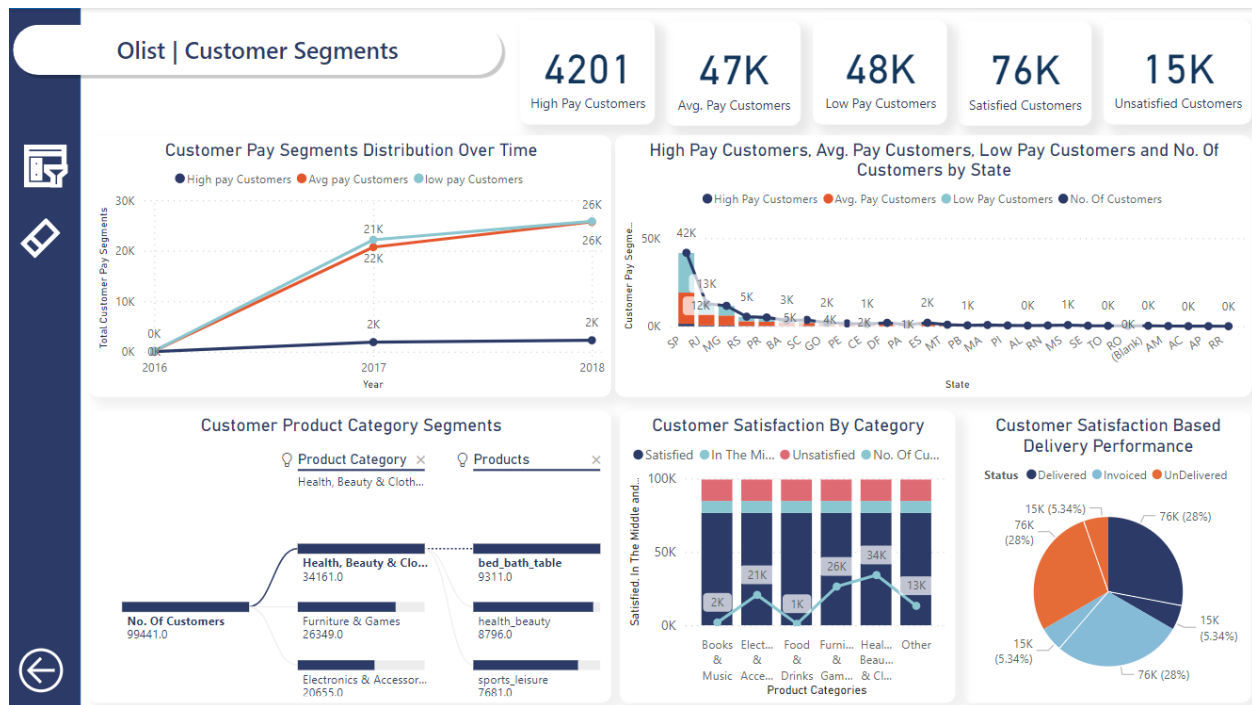


- Home
- Overview
- Customer Segments
- Orders Report
- Products Report
- Distance Map

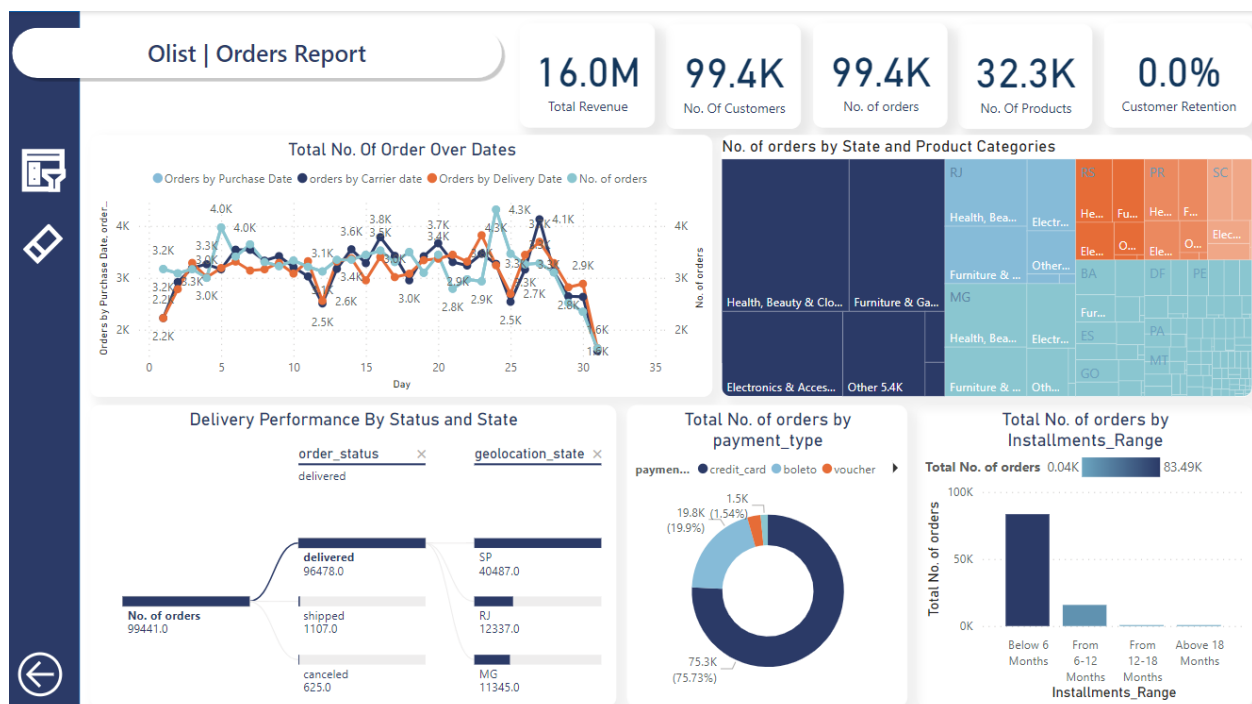
Home Page



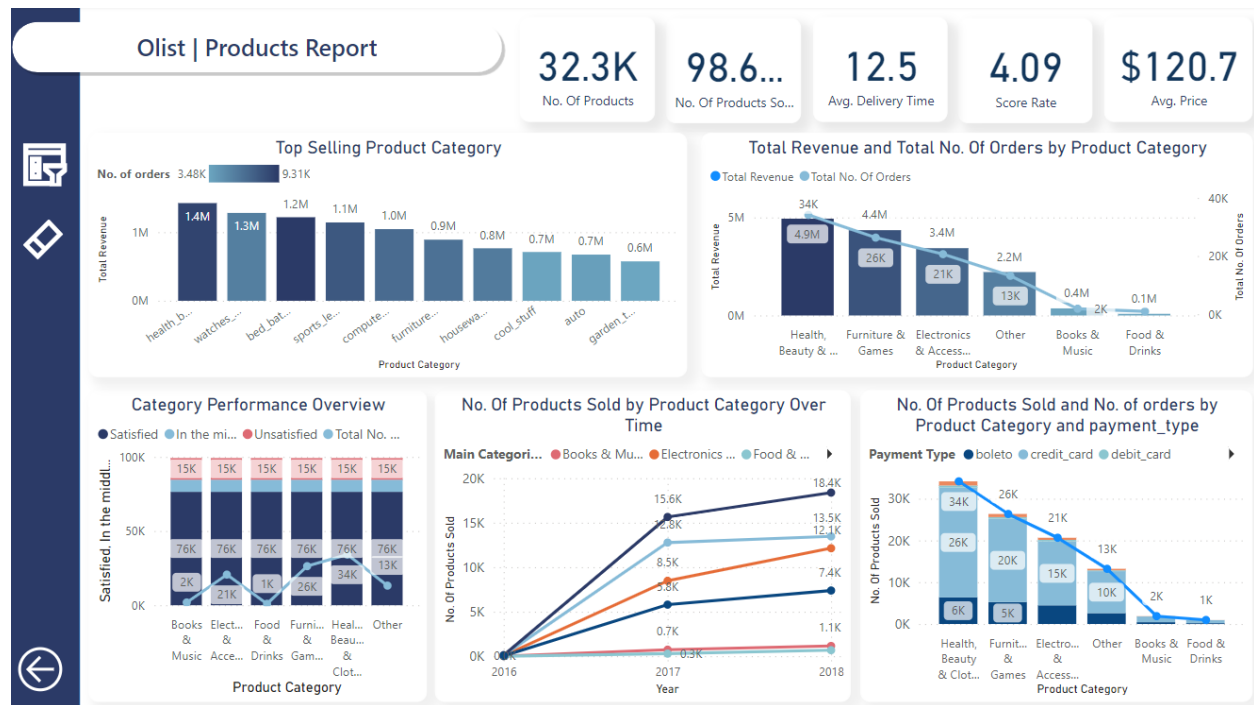
Overview



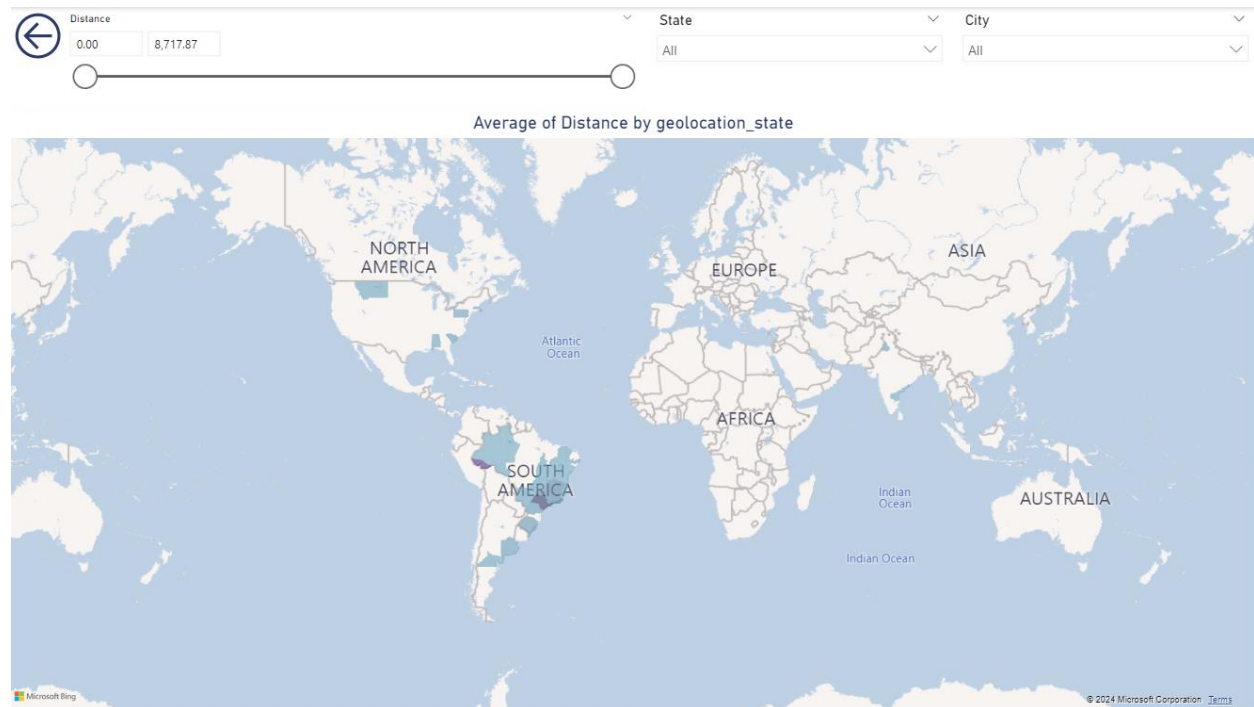
Customer Segmentation Report



Orders Report



Product report



Map shows the difference between sellers and customers